
AUTOMATING CUSTOMER SERVICE QA

A Modular, GenAI-Powered Evaluation & Reporting Pipeline

Built using LangChain Expression Language (LCEL), Pydantic validation, and multi-LLM (OpenAI & Gemini) routing.

The Customer Support QA Challenge

The Status Quo

Manual Auditing

Managers spend hours listening to calls, creating immense operational overhead.

1-2% Sampling Size

QA checks miss critical billing claims, customer complaints, and compliance risks.

Subjective Feedback

Evaluation scoring varies between managers, leaving agents with inconsistent training inputs.

The Automated Vision

100% Audit Coverage

Generative AI automatically processes every transcript instantly, missing zero details.

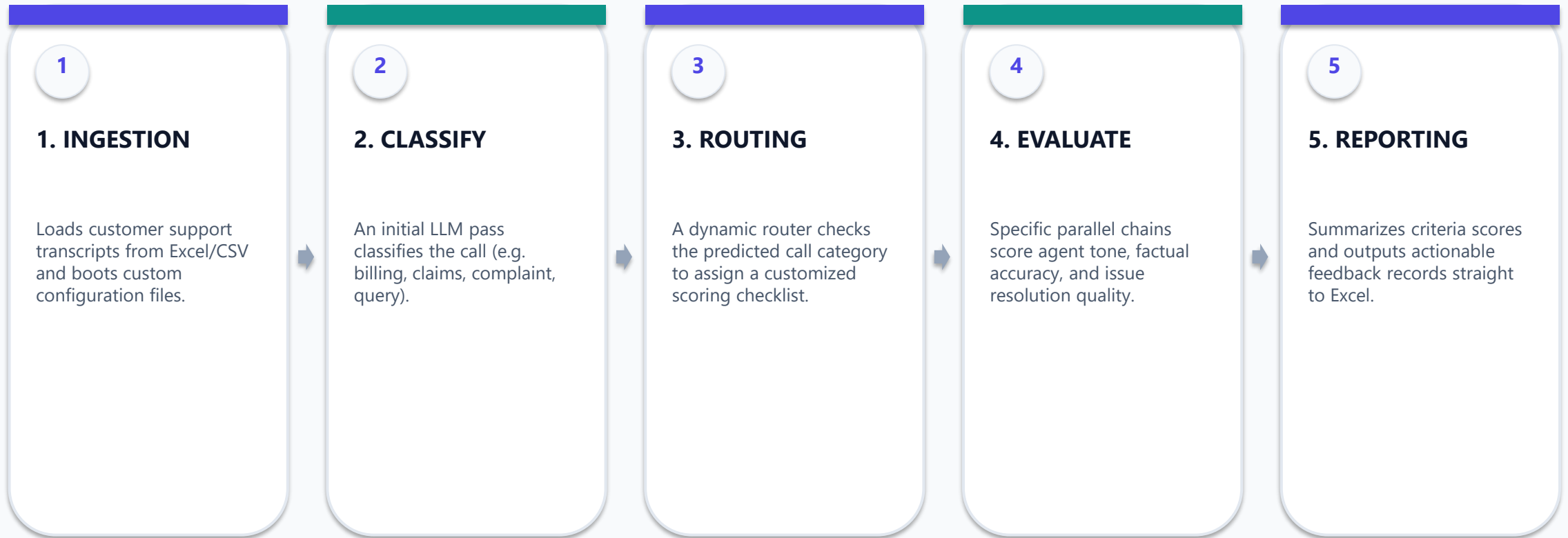
Objective Metrics

Every call is evaluated against unified prompts for Tone, Empathy, Knowledge, and Resolution.

Continuous Feedback Loop

Standardized spreadsheets log agent ratings and specific training actions automatically.

System Architecture & Execution Flow



Smart Routing: Token & Context Optimization

Why Route Dynamically?

Running every evaluator on every single call transcript is highly wasteful and injects noise.

- Billing calls do not benefit from a detailed empathy audit.
- Complex complaints require sensitive tone monitoring.
- Simple FAQs only require knowledge accuracy verification.

By classifying first and routing transcripts dynamically, the system reduces LLM token costs by up to 50% and improves the relevancy of QA scores.

Category Scoring Configurations



Billing & Claims Inquiry

Knowledge Accuracy + Issue Resolution Quality



Customer Complaint

Tone & Empathy + Issue Resolution Quality



General Queries / FAQ

Knowledge Accuracy (Factual Precision Only)



Fallback / Others

Knowledge Accuracy (Default Safety Catch)

Engineered for Enterprise Reliability

LangChain Expression Language (LCEL)

The pipeline is assembled using LCEL's declarative syntax:

```
`chain = prompt | llm | parser`
```

This design enables:

- Seamless custom template bindings.
- Easy parallel pipeline branches.
- Structured fallbacks and retry behaviors.

Pydantic Structured Outputs

Relying on LLM text output is highly fragile in production.

- The system binds a strict schema via Pydantic BaseModel.
- Generates structured JSON outputs representing scores, categories, and justifications.
- Eliminates regex cleaning or string parsing code completely.

Multi-Provider Agility

Infrastructure flexibility is built directly into the core code.

- Supported providers: OpenAI (GPT-4o-mini) and Google Gemini (Gemini 1.5 Flash).
- Configuration changes are fully decoupled from logic via config.json.
- Facilitates cost-optimization and local LLM migrations.

Actionable Reporting & Excel Integration

The Manager Report Chain

Once evaluation scores (1-5) and reasoning justifications are generated, the system passes the logs to a Final Executive QA Manager Chain.

This component distills raw evaluations into:

1. Performance Summary: A professional, context-rich performance report of the agent's work.
2. Actionable Recommendations: A clear, bulleted checklist detailing training points for the agent.

This translates metric raw scores directly into operational value.

Batch Processing & Data Export

The entire dataset is consolidated and exported to an Excel sheet (data/output.xlsx) containing:

- Predicted Call Categories & Confidences.
- Detailed Criteria Scores (Empathy, Accuracy, Resolution).
- Written Manager Summaries & Actionable Recommendations.

 Error Tolerant design: The pipeline includes explicit file-lock catch boundaries (e.g. checking for Windows Excel lock errors) to avoid data corruption or write failures.

SUMMARY

Operational Benefits & Outcomes

100% Quality Auditing

Enables auditing of all customer support logs, eliminating sampling errors completely.

Standardized Agent Performance

Unifies grading criteria to ensure feedback transparency and build agent trust.

Token & API Savings

Smart classification and routing avoids running wasteful chains on simple inquiries.

Provider Independence

Decoupled architecture makes it simple to swap model vendors (OpenAI/Gemini/Local LLMs).